

COMPACT MODULI OF ENRIQUES SURFACES WITH A NUMERICAL POLARIZATION OF DEGREE 2

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ABSTRACT. We describe a geometric, stable pair compactification of the moduli space of Enriques surfaces with a numerical polarization of degree 2, and identify it with a semitoroidal compactification of the period space.

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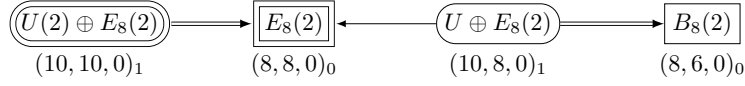
1. INTRODUCTION

Enriques surfaces are quotients of K3 surfaces by basepoint free involutions. They satisfy $2K \sim 0$ and $q = 0$ and occupy a place somewhere in between rational and K3 surfaces. Unlike K3 surfaces, there are only finitely many moduli spaces of polarized Enriques surfaces, see [GH16]. Each of them parameterizes the same surfaces, with some finite data attached.

In this paper we consider the moduli space $F_{\text{En},2}$ of pairs $(Z, [\mathcal{L}_Z])$, where Z is an Enriques surface with ADE singularities and $[\mathcal{L}_Z] \in \text{Pic } Z/\mathbb{Z}_2$ is an ample numerical polarization of degree 2. Equivalently, this is the moduli space of ADE Enriques surfaces $(Z, \mathcal{M})$ with a 2-divisible polarization $\mathcal{M} = \mathcal{L}_Z^{\otimes 2} \in \text{Pic } Z$ of degree 8. The Baily-Borel compactification $\overline{F}_{\text{En},2}^{\text{BB}}$ of this space was described by Sterk [Ste91].

By the classification of big and nef linear system on Enriques surfaces [Cos83], see also [CD89, CDL24], the linear system $|\mathcal{M}|$ is basepoint free and defines a double cover $\rho: Z \rightarrow W$ to a quartic del Pezzo surface with $4A_1$ or $A_3 + 2A_1$ singularities. The ramification divisor $R_Z \in |\mathcal{M}|$ of ρ is ample and the pair $(Z, \epsilon R_Z)$ is log canonical for any $0 < \epsilon \ll 1$. Thus, the moduli space $F_{\text{En},2}$ admits a geometric, modular compactification $\overline{F}_{\text{En},2}$ in the KSBA moduli space of pairs $(Z, \epsilon R_Z)$ with semi log canonical singularities, $K_Z \equiv 0$ and ample $\mathbb{Q}$ -Cartier divisor R_Z . See [Kol23] for their general theory. Our main result, in Section 5.2, is

Date: December 27, 2023.

FIGURE 3. Cusp diagram of $F_{(10,10,0)}$, for $T_{\text{En}} = U \oplus U(2) \oplus E_8(2)$

A geometric interpretation of these cusps is as follows. Let $\mathcal{X} \rightarrow (C, 0)$ be a Kulikov model and consider the completed period mapping

$$(C, 0) \rightarrow \overline{F}_{(10,10,0)}^{\text{BB}}.$$

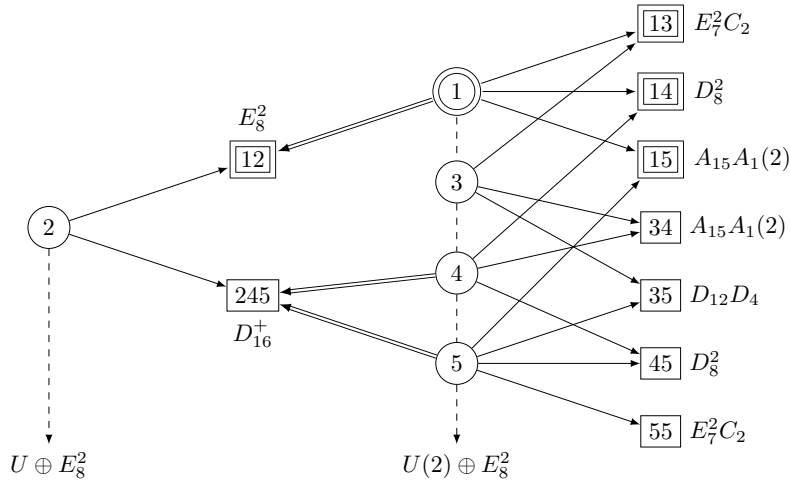
Suppose that $\iota_{\text{En},t}$ on the generic fiber extends to an involution $\iota_{\text{En},0}$ on the central fiber. If $0 \in C$ is sent to the double-circled cusp $(10, 10, 0)_1$ or $(8, 8, 0)_0$ then $\iota_{\text{En},0}$ is basepoint free. Otherwise, $\iota_{\text{En},0}$ has a nonempty finite set of fixed points.

Furthermore, the dual complex $\Gamma(\mathcal{X}_0)$ is a 2-sphere and the induced action of $\iota_{\text{En},0}$ on $\Gamma(\mathcal{X}_0)$ in the $(10, 10, 0)_1$ case is an antipodal involution, while in the $(10, 8, 0)_1$ case it is a hemispherical involution, see [AE22, Sec. 8F]. So the quotients of $\Gamma(\mathcal{X}_0)$ by the Enriques involution are, respectively, the real projective plane $\mathbb{RP}^2$ and a disk $\mathbb{D}^2$. In Type II, $\Gamma(\mathcal{X}_0)$ is a segment. In the $(8, 8, 0)_0$ case, the action of $\iota_{\text{En},0}$ flips the segment, whereas in the $(8, 6, 0)_0$ case it fixes the segment.

3.3. Cusp diagram of $F_{\text{En},2}$. Sterk [Ste91] computed the cusp diagram for $\overline{F}_{\text{En},2}^{\text{BB}}$. There are five 0-cusps for which we use Sterk's numbering 1, 2, 3, 4, 5. There are also 9 distinct 1-cusps.

Notation 3.1. We denote a 1-cusp by $i_1 \dots i_k$ if its closure contains the 0-cusps $i_1, \dots, i_k$. Here, $1 \leq k \leq 5$.

Lemma 3.2. *The morphisms $F_{\text{En},2} \rightarrow F_{(2,2,0)}$ and $F_{\text{En},2} \rightarrow F_{(10,10,0)}$ extend to the Baily-Borel compactifications, mapping 0-cusps to 0-cusps and 1-cusps to 1-cusps in the manner shown in Figure 4.*

FIGURE 4. Cusps of $\overline{F}_{\text{En},2}^{\text{BB}}$ with images in $\overline{F}_{(10,10,0)}^{\text{BB}}$ and $\overline{F}_{(2,2,0)}^{\text{BB}}$